Sample title

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2026

Basic concepts

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- A canonical example is
- However, not all matroids can be put into this form! Matroids that can be represented as such with a matrix with entries in $\mathbb{F}$ are called $\mathbb{F}$-representable.
- We need to introduce some concepts in order to state the problem. We would omit all proofs due to time constraint.

- Analogous concepts from linear algebra still carry over:
  - basis: maximal independent set
  - rank: still true that all bases have the same size, called the rank
- We often describe matroids using a labelled set and set of basis, e.g. $E = \{0, 1, 2\}$, $\mathcal{B} = \{\{1, 2\}, \{0, 2\}\}$.
- There is a notion of duality:
- Two relevant fundamental operations:
  - Deletion:
  - Contraction: for a subset $T \subset E$, contraction $M/T = (M^* \setminus T)^*$.

Problem statement

- The problem we solved: classify all rank 4, 11-element, excluded minor for $\mathbb{F}_5$ representability.
- We call a $\mathbb{F}_5$-representable matroid a *quinary* matroid.
- An *excluded minor for* $\mathbb{F}_5$ *representability* is a matroid M which is not quinary, but for all $e \in E(M)$, both M/e and $M \setminus e$ are quinary.
- These are in some sense, the 'minimal' non-quinary matroids.
- The study of excluded minors is extremely important in matroid theory.

Comparison of methodology

- We used LLM to solve the problem.
- However, directly asking the question does not lead to full resolution.
- Brute-force using computational tools is needed! Placing LLM in an equipped environment and asking it to use those tools leads to a complete resolution.

Proof outline

- The proof can be broken down into the a few steps.
- The broad idea is to view a 11-element excluded minor M as an extension of a 10-element quinary matroid $N = M \setminus e$.
- For each N, we want extensions M which are non-quinary but all of whose deletions and contractions are quinary.
- First generate all appropriate N.
- Then exhaust all possible M which are non-quinary but all of whose deletions are quinary. This is done by listing all appropriate quinary extensions (by e) of N and $N \setminus f$ for all $f \in E$.
- Finally, pick those whose all contractions are quinary. Then deduplicate with respect to isomorphism.

We shall give a bit more details for the steps, especially the computations involved.

Step 1

1. Generate 10-element quinary matroids. Due to a key result, we only need connected simple rank-4 ones up to isomorphism. Tool used for generation: SageMath.

Step 2

Step 3

Future work

We plan to settle the classification without the condition of rank 4. By duality consideration and existing work, the only case left is rank 5.